KADI SARVA VISHWAVIDHYALAYA

B.E. Semester IV EXAMINATION (April - 2025)

Subject Code: CT405-N Subject Name: OBJECT ORIENTED PROGRAMMING USING JAVA
Date: 30/04/2025 Time: 10:00 AM TO 01:00 PM Total Marks: 70 Marks

Instructions:
1. **All questions are compulsory.**
2. **Figures to the right indicate full marks.**
3. **Use of scientific calculator is permitted.**
4. **Indicate clearly, the options you attempt along with its respective question number.**
5. **Use the last page of main supplementary for rough work.**

Section-I

Q:1 Attempt Following.

(A) Explain the different types of control structures in Java with examples. [5]

(B) What is method overloading? Explain with an example. [5]

(C) Discuss the various types of arrays in Java with examples. [5]

OR

Q:1 (C) Write a Java program to demonstrate the use of the Math class functions such as sqrt(), pow(), and abs(). [5]

Q:2 Answer the following question.

(A) Explain constructors and finalizers with examples. [5]

(B) What is the difference between String and StringBuffer classes? Give examples. [5]

OR

Q:2 (A) Write a Java program to demonstrate the use of the 'this' reference. [5]

(B) Explain the visibility modifiers in Java with suitable examples. [5]

Q:3 Answer the following question.

(A) What is polymorphism? Explain dynamic binding with an example. [5]

(B) Differentiate between interfaces and abstract classes in Java.. [5]

OR

Q:3 (A) Write a Java program to demonstrate the use of the 'instanceof' operator. [5]

(B) Explain packages in Java. Discuss the UTIL package with an example. [5]

Section II

Q:4 (All Compulsory)

(A) Explain exception handling in Java using try-catch-finally with an example. [5]

(B) What is thread synchronization? Explain with an example. [5]

(C) Explain the thread life cycle in Java with a suitable diagram. [5]

OR

Q:4 (C) Write a Java program to create a thread using the Runnable interface and display a message from the thread. [5]

Q:5 Answer the following question.

(A) Explain the difference between Text I/O and Binary I/O in Java. [5]

(B) What is event handling in Java? Explain different types of events with examples. [5]

OR

Q:5 (A) Write a Java program to demonstrate Random Access Files in Java. [5]

(B) Explain Object I/O with an example. [5]

Q:6 Answer the following question.

(A) What are Layout Managers in Java? Explain different types with examples. [5]

(B) Explain GUI components in Java with an example of any three components. [5]

OR

Q:6 (A) Write a Java program to create an applet that displays "Hello, Java!" and changes color on mouse click. [5]

(B) What are menus and dialog boxes in Java? Explain with an example. [5]

Exam Number: _____

KADI SARVA VISHWAVIDYALAYA

B.E. SEMESTER – IV (NEW) EXAMINATION OCTOBER-2024

Subject Name: Object Oriented Programming Using Java **Subject Code: CT405-N**

Date: 23/10/2024 (Wednesday) **Time: 12:30 p.m. to 03:30 p.m.** **Total Marks: 70**

Instructions:

1. **All questions are compulsory.**
2. **Figures to the right indicate full marks.**
3. **Indicate clearly, the options you attempt along with its respective question number.**
4. **Use the last page of main supplementary for rough work.**

Section-I

Q.1	(A)	Explain Java garbage collection mechanism.	[5]
	(B)	List OOP characteristics and describe inheritance with examples.	[5]
	(C)	Distinguish between method overloading and overriding with suitable example.	[5]
		OR	
	(C)	Write a program which checks whether entered number is prime or not.	[5]
Q.2	(A)	Explain following keywords: this and super with example.	[5]
	(B)	Write a program to find length of string and print second half of the string.	[5]
		OR	
Q.2	(A)	Explain different types of constructor with example.	[5]
	(B)	Explain different access modifiers with example.	[5]
Q.3	(A)	Differentiate Abstract class and Interface with suitable example.	[5]
	(B)	Define ArrayList class. Write a program to add three elements in ArrayList.	[5]
		OR	
Q.3	(A)	Differentiate String with StringBuffer class. List out the methods available with String class and explain any two with proper JAVA code in detail.	[5]
	(B)	Explain Instance of operator with example.	[5]

Section-II

Q.4	(A)	What is an Exception? List out various built-in exceptions in JAVA and Explain any one Exception class with suitable example.	[5]
	(B)	What is Multithreading? Explain Thread life cycle with proper JAVA code.	[5]
	(C)	The abstract Vegetable class has three subclasses named Potato, Brinjal and Tomato. Write an application that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of Object to return a string with the name of the vegetable and its color.	

OR

	(C)	Write a program that takes the name of a text file as argument and prints every line in lower case.	[5]
Q.5	(A)	What is synchronization? Write a program that uses thread synchronization with proper output.	[5]
	(B)	Write a program to insert values in array and generate exception for index out out of range.	[5]

OR

Q.5	(A)	Explain event handling in java with example.	[5]
	(B)	Explain Dynamic method dispatch in java with example.	[5]
Q.6	(A)	Explain different Layouts in Java with example.	[5]
	(B)	Write a program to demonstrate the creation of custom exception and throw statement	[5]

OR

Q.6	(A)	Explain Character I/O classes with example.	[5]
	(B)	Explain life cycle of Applet .	[5]

KADI SARVA VISHWAVIDHYALAYA

B.E Semester-IV Examination (CE / IT / CSE) April 2024

Subject Name: Object Oriented Programming Using Java

Subject Code: CT405-N

Date: 09/05/2024 **Time: 12.00 P.M. TO 3.00 P.M.** **Total Marks: 70**

Instructions:

1. **Answer each section in a separate answer sheet.**
2. **All questions are Compulsory.**
3. **Indicate clearly, the option you attempt along with its respective question number.**
4. **Use the last page of the main supplementary of rough work.**

Section-I

Q-1 (A) What is Byte Code? Interpret the different states of Java Program execution? [5]

(B) Define Method Overloading & list its types. Create a class named 'PrintNumber' to print various numbers of different datatypes by creating different methods with the same name 'printn' having a parameter for each datatype. [5]

(C) Define ArrayList in Collection interface. Illustrate working of ArrayList in a program with add() and remove() methods. Add 5 elements and remove the 3rd element. [5]

OR

(C) Define LinkedList in Collection interface. Illustrate working of LinkedList in a program with add() and remove() methods. Add 3 elements and remove the 1st element. [5]

Q-2 (A) Define Constructor. Discuss **Parameterized Constructor** with suitable code. [5]

(B) Explain the importance of **"this"** keyword with proper examples. [5]

OR

Q-2 (A) Write a program to print the names of students by creating a Student class. If no name is passed while creating an object of Student class, then the name should be "Unknown", otherwise the name should be equal to the String value passed while creating object of Student class. Use the concept of Constructor Overloading. [5]

(B) Explain Visibility Modifiers with examples. [5]

Q-3 (A) Explain Dynamic Method Dispatch with a proper example. [5]

(B) Define Inheritance. Discuss in brief types of Class Inheritance with codes. [5]

OR

Q-3 (A) Define Interface. Discuss Partial Implementations with examples. [5]

(B) Discuss the concept of Abstract Class with proper example. [5]

Section-II

Q-4	(A)	Explain Lifecycle of Thread with proper diagram and its methods.	[5]
	(B)	Explain Thread Synchronization with proper code.	[5]
	(C)	Explain try-catch-finally block with example.	[5]
		OR	
	(C)	Explain Custom Exception with proper examples.	[5]
Q-5	(A)	Explain in brief Byte Stream Classes in java.io.	[5]
	(B)	Write a program to read characters from the console using BufferedReader.	[5]
		OR	
Q-5	(A)	Explain in brief Character Stream I/O Classes in java.io.	[5]
	(B)	Write a program to read a string from the console using BufferedReader..	[5]
Q-6	(A)	Define applet. Discuss its lifecycle.	[5]
	(B)	Explain any 2 GUI Components.	[5]
		OR	
Q-6	(A)	Discuss in brief Delegation Event Model in java.	[5]
	(B)	Discuss in brief Layout Managers.	[5]

Exam Number: _____

KADI SARVA VISHWAVIDYALAYA

B.E. SEMESTER 4th (ATKT) EXAMINATION OCTOBER-2023

Subject Name: Object Oriented Programming using Java **Subject Code: CT405-N**
Date: 04/11/2023 (Saturday) **Time: 12:00 p.m. to 03:00 p.m.** **Total Marks: 70**

Instructions:

1. **All questions are compulsory.**
2. **Figures to the right indicate full marks.**
3. **Indicate clearly, the options you attempt along with its respective question number.**
4. **Use the last page of main supplementary for rough work.**

Section-I

Q.1	(A)	Discuss significance of byte code.	[5]
	(B)	List OOP characteristics and describe inheritance with examples	[5]
	(C)	List and explain various features of Java.	[5]
		OR	
	(C)	Explain all,access modifiers and their visibility as class members.	[5]
Q.2	(A)	Compare String with StringBuffer. Also, write a program to count the occurrence of a character in a string.	[5]
	(B)	How can you create package in Java? Explain with example.	[5]
		OR	
Q.2	(A)	Explain the words **final** and **this** with the help of an example.	[5]
	(B)	Explain types of constructor with the help of an example	[5]
Q.3	(A)	What is collection in Java? Differentiate between Vector and ArrayList.	[5]
	(B)	Write a JAVA program for matrix multiplication.	[5]
		OR	
Q.3	(A)	Explain the following: i) Arguments and Parameters of a function ii) Pass by Value and Pass by reference	[5]
	(B)	Compare abstract & interface in Java.	[5]

Section-II

Q.4	(A)	What is an Exception? Explain try, catch and finally with example.	[5]
	(B)	What is thread? Describe the complete life cycle of thread	[5]
	(C)	What is the keyword "throw" and "throws" used for?	[5]
		OR	
	(C)	Write a program that illustrates interface inheritance. Interface P is extended by P1 and P2. Interface P12 inherits from both P1 and P2. Each interface declares one constant and one method. class Q implements P12. Instantiate Q and invoke each of its methods. Each method displays one of the constants.	[5]
Q.5	(A)	Compare Binary I/O and text I/O.	[5]
	(B)	Explain Thread Synchronization with example.	[5]
		OR	
Q.5	(A)	Differentiate Application and Applet.	[5]
	(B)	Explain event handling in Java with suitable example.	[5]
Q.6	(A)	Explain the various Character I/O classes with example.	[5]
	(B)	Explain various layouts in Java with proper example.	[5]
		OR	
Q.6	(A)	Write a program to explain the concept of TreeSet.	[5]
	(B)	Explain the lifecycle of Applet.	[5]

Exam Number: _____

KADI SARVA VISHWAVIDYALAYA

B.E. SEMESTER 4th (REG/ATKT) EXAMINATION MAY-2023

Subject Name: Object Oriented Programming using Java **Subject Code: CT405-N**

Date: 15/05/2023 (Wednesday) **Time: 10:00 a.m. to 01:00 p.m.** **Total Marks: 70**

Instructions:

1. **All questions are compulsory.**
2. **Figures to the right indicate full marks.**
3. **Indicate clearly, the options you attempt along with its respective question number.**
4. **Use the last page of main supplementary for rough work.**

Section-I

Q.1 (A) List and explain the features of Java. [5]

(B) List the various operators in Java and explain any two with proper example. [5]

(C) What is control structures and selection in Java? Explain it with example. [5]

OR

(C) Write a JAVA program to implement the Fibonacci series using for loop control structure. [5]

Q.2 (A) Discuss various access modifiers available in JAVA? How access modifier affects the visibility of a member in different access locations? Explain with suitable example. [5]

(B) Explain constructor in java along with its rules and types. Also write down the program to demonstrate the concept of constructor overloading. [5]

OR

Q.2 (A) Explain the given keywords with proper example. 1) this 2) ~~final~~ super [5]

(B) Write a JAVA program for matrix multiplication. [5]

Q.3 (A) Differentiate method overloading and method overriding with the help of example. [5]

(B) What is collection in Java? Differentiate between Vector and ArrayList. [5]

OR

Q.3 (A) Explain the use of final keyword in JAVA with proper example. [5]

(B) Compare abstract & interface in Java. [5]

Explain the Process of thread Creation with example.

Section-II

Q.4	(A)	What is Inheritance? List out its types and explain any one with proper example.	[5]
	(B)	What is Polymorphism? List out its types and explain method overriding with program.	[5]
	(C)	Compare Binary I/O and text I/O.	[5]

OR

	(C)	Describe abstract class called Shape which has three subclasses say Triangle, Rectangle, and Circle. Define one method area() in the abstract class and override this area() in these three subclasses to calculate for specific object i.e. area() of Triangle subclass should calculate area of triangle etc. Same for Rectangle and Circle.	[5]
Q.5	(A)	What is an Exception? Explain try, catch and finally with example	[5]
	(B)	What is multithreading? Explain thread life cycle in java.	[5]

OR

Q.5	(A)	Write a program to raise and handle divide by zero exception.	[5]
	(B)	Explain synchronization in thread with example.	[5]
Q.6	(A)	Explain the lifecycle of Applet.	[5]
	(B)	Explain event handling in Applet with example.	[5]

OR

Q.6	(A)	Explain the various Character I/O classes with example.	[5]
	(B)	Differentiate Application and Applet.	[5]

Seat No:- _____

KADI SARVA VISHWAVIDHYALAYA

B.E. CE/IT/CSE Semester 4 Examination- November-2022

Subject Code:- CT405-N **Subject Name:- OOPJ**

Time:- 10.00 AM to 01.00 PM

Date: 14/11/2022 **Total Marks:- 70**

Instructions:

1. Answer each section in separate Answer sheet.
2. All questions are **Compulsory**.
3. Indicate **clearly**, the options you attempt along with its respective question number.
4. Make suitable **assumption** where needed.

Section – I

Q-1 **Answer the following questions.** (All compulsory)

(A) List and explain various features of java. [5]

(B) What is constructor? Explain constructor overloading of java with example. [5]

(C) Differentiate between Method overloading and Method overriding of java. [5]

OR

(C) Explain any two methods of String Class and String Buffer Class of java with suitable example. [5]

Q-2 **Answer the following questions.**

(A) Explain this and static keywords of java with suitable example. [5]

(B) Explain use of protected and default access specifiers of java with suitable example. [5]

OR

Q-2 (A) Write a java program to demonstrate abstract class with suitable example. [5]

(B) Write a java program of multilevel inheritance with suitable example. [5]

Q-3 **Answer the following questions.**

(A) Explain super and final keywords of java with suitable example [5]

(B) Explain dynamic method dispatch of java with suitable example? [5]

OR

Q-3 (A) Write a java program to implement Arraylist and Treeset classes. [5]

(B) Write a java program to implement user define package. [5]

Section – II

Q-4 **Answer the following questions.** (All compulsory)

(A) Explain use of try, catch, finally with suitable example in exception handling of java. [5]

(B) Explain life cycle of thread in java. [5]

(C) Write a java program to copy content of file1.txt to file2.txt using java file handling. [5]

OR

Q-4 (C) Write a java programs to handle Arithmetic Exception. [5]

Q-5 **Answer the following questions.**

(A) Explain Java Border Layout with suitable example. [5]

(B) Write a java programs to handle ArrayIndexOutOfBounds Exception. [5]

OR

Q-5 (A) How can we open and read a text file in java? Explain your answer with example. [5]

(B) Write a java program to implement custom exception (user define exception). [5]

Q-6 **Answer the following questions.**

(A) Write a java program to define two threads one will print 1 to 10 numbers whereas other will print 11 to 20 numbers. [5]

(B) Explain Java Grid Layout with suitable example. [5]

OR

Q-6 (A) Explain life cycle of Applet in java. [5]

(B) Give the output of following java code: [5]

```
public class Code{

  public static void main(String args[])
  {
    StringBuffer str1 = new StringBuffer("ldrp");
    StringBuffer str2 = str1;
    str1.append("itr");
    System.out.println(str1 + " " + str2 + " " + (str1 == str2));
  }
}
```

Enrollment No_____

KADI SARVA VISHWAVIDYALAYA

B.E. SEMESTER – IV REGULAR EXAMINATION June -2022

SUBJECT CODE: - CT405-N **SUBJECT NAME:- Object Oriented Programming Using Java**

DATE: -15-06-2022 **TIME: - 12:30 TO 3:30PM** **MARKS:-70**

Instructions:

1. Answer each section in separate Answer Sheet.
2. All questions are **compulsory**.
3. Indicate clearly, the options you attempted along with its respective question number.
4. Assume suitable data wherever necessary.
5. Use of scientific calculator is permitted.

SECTION-I

Q-1 (A) List and Explain features of Java . **[05]**

(B) What is byte code? Justify why Java is platform independent . **[05]**

(C) Compare Object Oriented programming with procedural programming language. **[05]**

OR

(C) Explain following keywords by example **[05]**
this, final ,super

Q-2 (A) Give output of following code. **[05]**
I . if the output of any program is an error according you, you have to mention which type of error it is. Compile time, Runtime and Justify.

```
public class ABC
{
  public static void main (String args[])
  {
    try
        {
            int a[]= new int[10];
            a[11]=9;
        }
        catch (ArrayIndexOutOfBoundsException e)
        {
            System.out.println("ArrayIndexOutofBounds");
        }
    }
}
```

```
II. class Ex
 {
     public static void main (String args[])
     {
      try
      {
          int a=10, int b=0;
          int output =a/b;
          System.out.println("Result:"+output);
      }
     catch(Arithmetic Exception e)
     {
          System.out.println("you Shouldn't divide number by zero");
     }
   }
}
```

(B) What is multithreading? Explain thread life cycle in java. [05]

OR

Q-2(A) Explain visibility modifiers with their scope in packages. [05]

(B) What is String class? Explain String class method with example. [05]

Q-3(A) Explain method overloading with program. [05]

(B) What is Inheritance? Explain multilevel inheritance with program. [05]

OR

Q-3(A) What is Polymorphism? Explain method overriding with program. [05]

(B) Explain constructor in java and its rule with program. [05]

SECTION-II

Q-4(A) Compare abstract and interface in Java [05]

(B) What is package in java? Explain step to create package with example. [05]

(C) Explain Synchronization with example [05]

OR

(C) Explain Lifecycle of Applet [05]

Q-5(A) Write a program to print half pyramid using *

```
*
**
***
****
*****
```

[05]

(B) Compare Binary I/O and Text I/O [05]

OR

Q-5(A) What is Layout? Explain various Layout manager in java [05]

(B) Write a program to print half pyramid using alphabet [05]

```
A
BB
CCC
DDDD
EEEEE
```

Q-6(A) What is collection in java? Differentiate between Vector and ArrayList [05]

(B) Differentiate Application and Applet. [05]

OR

Q (A) Explain Event Handling in Applet with example. [05]

(B) abstract Vegetable class has three subclasses named Potato, Brinjal and Tomato. [05]
Write an application that demonstrates how to establish this class hierarchy.
Declare one instance variable of type String that indicates the color of a vegetable.
Create and display instances of these objects.

Best of Luck